

NAJRAN, Saudi Arabia

WMO: 411280

Lat: 17.611N

Lon: 44.419E

Elev: 1214

StdP: 87.57

Time Zone: 3.00 (E03)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.2	9.0	-12.2	1.5	18.1	-10.0	1.8	19.5	6.2	26.2	5.4	24.3	0.3	240	0.380

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	40.2	17.3	39.8	17.3	39.0	17.2	21.0	33.2	20.1	32.8	19.5	32.5	3.7	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	14.8	24.6	16.2	13.4	24.6	15.1	12.5	25.4	66.7	32.9	63.6	33.1	61.0	31.6	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.5	5.5	5.0	DB	4.6	41.6	1.7	0.8	3.4	42.1	2.4	42.6	1.4	43.0	0.1	43.5	(4)
(5)			WB	-0.2	22.0	1.8	1.6	-1.5	23.1	-2.6	24.1	-3.6	25.0	-5.0	26.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	25.8	18.0	20.9	24.1	26.9	29.7	31.4	33.0	32.4	29.4	24.5	20.9	18.1	(6)	
	DBStd	5.74	3.11	3.34	3.06	2.32	1.74	1.46	1.35	1.41	1.93	1.96	2.09	2.69	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	101	46	13	2	0	0	0	0	0	0	0	3	37	(9)	
	CDD10.0	5768	248	304	438	506	611	643	714	696	582	448	327	252	(10)	
	CDD18.3	2828	35	84	181	257	352	393	456	437	332	190	81	30	(11)	
	CDH23.3	41844	523	1071	2114	3247	5243	6358	7527	7087	4923	2406	907	439	(12)	
	CDH26.7	25206	165	457	1013	1762	3241	4212	5109	4726	3028	1129	254	109	(13)	
(14) Wind	WSAvg	2.1	1.6	1.9	2.2	2.3	2.2	2.3	2.8	2.6	2.1	1.7	1.5	1.4	(14)	
(15) Precipitation	PrecAvg	72	4	1	16	19	8	4	3	9	0	6	2	1	(15)	
	PrecMax	264	47	4	163	92	53	75	26	63	2	85	27	5	(16)	
	PrecMin	14	0	0	0	0	0	0	0	0	0	0	0	0	(17)	
	PrecStd	52	12	1	34	24	14	14	5	15	1	17	6	1	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.2	35.1	37.0	37.6	39.2	41.0	41.1	41.0	39.9	35.2	32.0	31.6	(19)	
		MCWB	15.0	15.3	17.0	16.6	16.6	17.0	17.6	17.8	17.1	15.5	14.9	15.9	(20)	
	2%	DB	30.8	33.2	35.2	36.2	38.2	40.0	40.2	40.0	38.2	34.0	30.7	29.3	(21)	
		MCWB	14.7	15.3	16.2	16.3	16.6	16.9	17.4	17.7	16.8	15.1	15.0	15.1	(22)	
	5%	DB	28.2	31.8	33.8	35.2	37.8	39.1	39.8	39.1	37.2	32.9	29.2	27.8	(23)	
		MCWB	14.2	15.1	16.0	16.3	16.6	16.7	17.3	17.7	16.5	14.9	14.5	14.1	(24)	
	10%	DB	26.2	29.8	32.0	34.2	36.9	38.2	38.9	38.5	36.2	31.9	28.1	25.9	(25)	
		MCWB	13.6	14.6	15.6	16.1	16.5	16.4	17.3	17.7	16.4	14.8	14.0	13.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.5	19.1	20.4	21.0	21.5	21.7	21.7	21.8	19.8	19.9	18.4	17.9	(27)	
		MCDB	27.2	28.4	30.8	31.0	34.9	33.6	35.2	33.4	34.2	25.2	27.3	27.9	(28)	
	2%	WB	16.6	17.3	19.0	19.7	19.8	20.0	20.3	20.6	18.8	18.5	17.2	16.2	(29)	
		MCDB	26.0	28.8	27.8	29.1	31.8	35.1	34.4	33.4	34.7	26.8	26.1	26.5	(30)	
	5%	WB	15.4	16.2	18.1	18.9	18.9	18.9	19.5	19.8	17.9	17.3	16.2	15.2	(31)	
		MCDB	25.2	28.4	28.0	28.7	31.2	34.5	34.4	34.0	34.5	28.4	25.4	25.4	(32)	
	10%	WB	14.2	15.2	17.2	18.1	18.1	17.8	18.7	19.1	17.1	16.0	15.2	14.2	(33)	
		MCDB	24.4	27.7	27.9	28.4	31.1	34.9	34.7	34.4	34.7	29.7	25.3	24.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	15.1	15.5	14.4	13.9	14.2	14.5	12.4	13.0	14.5	15.0	14.2	14.8	(35)	
		MCDBR	18.2	18.2	16.6	15.6	15.3	15.0	13.2	13.5	14.7	15.5	15.6	16.9	(36)	
	5% WB	MCWBR	8.2	7.7	6.5	5.8	5.6	5.6	4.8	4.9	5.7	6.4	6.7	7.5	(37)	
		MCWBR	15.3	15.6	13.4	12.9	12.7	13.6	11.7	12.6	13.9	12.6	12.8	14.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.390	0.431	0.493	0.511	0.549	0.614	0.669	0.637	0.530	0.409	0.393	0.374	(40)	
		taud	2.161	2.020	1.859	1.846	1.741	1.588	1.539	1.592	1.768	2.075	2.134	2.190	(41)	
	Ebn at Noon		888	868	825	811	771	714	678	705	782	878	877	891	(42)	
		Edh at Noon	142	172	209	213	234	270	284	271	226	161	145	134	(43)	
(44) All-Sky Solar Radiation	RadAvg		5.38	6.11	6.77	7.09	7.47	7.43	6.83	6.77	6.93	6.60	5.68	5.25	(44)	
	RadStd		0.24	0.23	0.27	0.33	0.26	0.23	0.24	0.24	0.13	0.15	0.19	0.24	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.33	N/A	N/A	+0.55	N/A	N/A	N/A	N/A	+174	+170
(47) Regional (2 neighbors)		+0.46	N/A	N/A	+0.59	N/A	N/A	N/A	-46	+188	+139

Nomenclature: See separate page